

Chemistry — Assignment 12: SN1 vs SN2

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Chemistry (Foundational Chemistry) · Semester 5

Question. Distinguish between SN1 and SN2 reactions.

Feature	SN1	SN2
rate law	$k[R-X]$ (first order)	$k[R-X][Nu]$ (second order)
steps	2 (via carbocation)	1 (concerted)
intermediate	carbocation	none
rearrangement	possible	none
favoured by	3° substrate, polar protic, weak Nu	1° substrate, polar aprotic, strong Nu
stereochemistry	racemisation	inversion (Walden)

Prepared by Anshuman Atrey · structures rendered with RDKit.